

DS	Session	Stack	Question Information	Level 1 Challenge 47
Question description You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined: $F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$ $G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$ Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i, output -1 as its answer. If such a number does exist, output $A[G(F(i))]$ Constraints: $1 \leq N \leq 3000$ $0 \leq A[i] \leq 10^{18}$ Input : The first line contains a single integer N denoting the size of array A. Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$. Output : Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or -1, if $G(F(i))$ does not exist.				

✓ Logical Test Cases

Test Case 1
INPUT (STDIN)
8 3 7 1 7 8 4 5 2
EXPECTED OUTPUT
1 4 4 4 -1 2 -1 -1

Test Case 2
INPUT (STDIN)
8 31 71 11 17 81 14 51 21
EXPECTED OUTPUT
11 14 14 14 -1 21 -1 -1

✓ Mandatory Test Cases

Test Case 1
KEYWORD
if(a[j]>a[i])

Test Case 2
KEYWORD
for(i=0;i<n;i++)

✓ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
9

Test Case 2
TOKEN COUNT
225

Test Case 3
NLOC
30